FRONT MATTER

1. **Title Page**
    - Project Title
    - Student Full Name & ID
    - Department & Institution
    - Supervisor Name & Title
    - Project Duration / Submission Date
2. **Certificate of Completion / Approval Page**
    - Signatures of:
        - Supervisor
        - Project Coordinator / Capstone Chair
        - Department Head
3. **Abstract (Engineering Summary)**
    - Problem
    - Proposed solution
    - Key components (hardware/software)
    - Test methods and main results
    - 200–300 words, written last
4. **Acknowledgements (optional)**
5. **Table of Contents**
6. **List of Figures**

7. **List of Tables**
8. **List of Symbols, Abbreviations & Acronyms**

MAIN REPORT BODY

1. Executive Summary

- Concise overview of the whole report
- Audience: engineering managers or non-specialists
- Includes objectives, methods, outcomes, and recommendations

2. Introduction

- Background and relevance
- Engineering context and motivation
- Problem Statement
- Objectives (General & Specific)
- Scope of the project
- Report structure outline

3. Requirements and Constraints

- Functional requirements
- Non-functional requirements
- Engineering constraints:
    - Cost
    - Power
    - Performance

    - Safety
    - Regulatory standards (if any)
- Success Criteria

4. System Design and Architecture

- System-level design description
- Block diagrams
- Major subsystems (e.g., sensors, controllers, interfaces)
- Hardware/software architecture
- Design choices and trade-offs

5. Materials and Methods

- Tools, components, platforms used
- Detailed description of:
    - Hardware (circuits, boards, sensors, PCBs)
    - Software (languages, libraries, frameworks)
- Engineering calculations (e.g., power, load, bandwidth)
- Simulation methods (if any)

6. Implementation

- Step-by-step development
- Schematic diagrams / PCB layouts
- Screenshots or UI layouts
- Source code organization

- Integration steps

7. Testing and Validation

- Test plan and strategy
- Unit tests, integration tests, system tests
- Test results and measurements
- Test tools (oscilloscopes, multimeters, simulation software)
- Performance analysis

8. Results and Discussion

- Present findings using graphs/tables
- Compare expected vs actual performance
- Failure cases and reasons
- Engineering insights

9. Conclusion and Recommendations

- Summary of what was built and tested
- Whether project met objectives
- Limitations
- Suggestions for improvement
- Opportunities for commercialization or further R&D

BACK MATTER

- **References**

    - IEEE style preferred
- **Appendices**
    - Datasheets
    - Full schematics
    - Code listings
    - Bill of Materials (BoM)
    - Gantt chart or timeline
    - Safety checks or test logs

ENGINEERING REPORT FORMATTING GUIDELINES

Element	Format
Paper size	A4
Font	Times New Roman, 12pt
Spacing	1.5 line spacing
Margin	1 inch on all sides
Section headings	Numbered (e.g., 1.0, 1.1, 2.0, etc.)
Figures & tables	Numbered and captioned
Units	SI units only
Citations	IEEE format in-line